

Google Colab / VS Code

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Python Data Generation

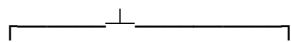
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Amazon S3

(Bronze / Silver / Gold)

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Glue Catalog Lake Formation

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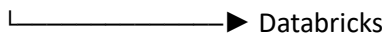
Athena

LF-Tags / IAM

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Unity Catalog

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Bronze → Silver → Gold

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Auto Loader → DLT → Workflow

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SQL Warehouse

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Dashboard

Resources Provisioned

- Amazon S3 (Data Lake)
- IAM Roles and Policies
- AWS Glue Crawlers
- AWS Glue ETL Jobs
- AWS Glue Data Catalog
- Amazon Athena
- AWS Lake Formation
- Amazon Redshift Serverless
- Databricks Workspace
- Unity Catalog
- Storage Credential
- External Location
- Bronze, Silver, and Gold Delta Tables
- Auto Loader
- Delta Live Tables (DLT) Pipeline
- Databricks Workflow
- SQL Warehouse
- Dashboard

Resources Torn Down

- Databricks Cluster
- SQL Warehouse
- Glue Crawlers
- Glue ETL Jobs
- Redshift Serverless
- Unused S3 resources (if applicable)

Monthly Cost at 10× Data Volume

If the data volume increased by 10×, storage costs would increase because more data would be stored in Amazon S3. Compute costs would also increase due to additional AWS Glue processing, more Athena queries, higher Redshift Serverless usage, and more Databricks cluster, DLT pipeline, and SQL Warehouse execution time. Overall, compute-related costs would grow more significantly than storage costs.